

EXPERIMENT NO. 7

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Class: T.Y B.Tech

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Subject: Advanced Machine
Learning

Department: AI & DS

Sem: VI

AIM: Implementation of Boosting

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[1]: # =====  
# AML PRACTICAL: BOOSTING (AdaBoost) WITH CROSS VALIDATION  
# =====  
  
# 1. Import Libraries  
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
  
from sklearn.datasets import load_breast_cancer  
from sklearn.model_selection import train_test_split, cross_val_score  
from sklearn.ensemble import AdaBoostClassifier  
from sklearn.tree import DecisionTreeClassifier  
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix  
  
# 2. Load Dataset  
data = load_breast_cancer()  
  
# Convert to DataFrame (for better understanding)  
df = pd.DataFrame(data.data, columns=data.feature_names)  
df['target'] = data.target  
  
print("Dataset Shape:", df.shape)  
print("\nFirst 5 rows:\n", df.head())  
  
# Features and Target  
X = df.drop('target', axis=1)  
y = df['target']  
  
# 3. Train-Test Split  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)  
  
print("\nTraining samples:", X_train.shape[0])  
print("Testing samples:", X_test.shape[0])  
  
# 4. Create AdaBoost Model  
# Weak Learner = Decision Stump (depth=1)  
base_model = DecisionTreeClassifier(max_depth=1)  
  
ada_model = AdaBoostClassifier(  
    estimator=base_model,  
    n_estimators=50,  
    learning_rate=1.0,  
    random_state=42  
)
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# 5. Train Model
ada_model.fit(X_train, y_train)

# 6. Predictions
y_pred = ada_model.predict(X_test)

# 7. Evaluation
print("\n===== MODEL EVALUATION =====")
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:\n")
print(confusion_matrix(y_test, y_pred))

# 8. Cross Validation (5-Fold)
print("\n===== CROSS VALIDATION =====")
cv_scores = cross_val_score(ada_model, X, y, cv=5, scoring='accuracy')

print("Fold Accuracies:", cv_scores)
print("Mean Accuracy:", np.mean(cv_scores))
print("Standard Deviation:", np.std(cv_scores))

# 9. Visualization of Cross Validation Scores
plt.figure(figsize=(6,4))
plt.plot(cv_scores, marker='o', linestyle='--', color='blue')
plt.title("Cross Validation Accuracy (5-Fold)")
plt.xlabel("Fold Number")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()

# 10. Hyperparameter Experiment (Optional for better marks)
print("\n===== PARAMETER TUNING =====")
estimators_list = [10, 50, 100, 200]

for n in estimators_list:
    temp_model = AdaBoostClassifier(
        estimator=DecisionTreeClassifier(max_depth=1),
        n_estimators=n,
        random_state=42
    )

    scores = cross_val_score(temp_model, X, y, cv=5)
    print(f"{n} estimators = {n}, Mean Accuracy = {scores.mean():.4f}")

# =====
# END OF PRACTICAL
# =====

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Output:

Dataset Shape: (569, 31)

First 5 rows:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	\
0	17.99	10.38	122.80	1001.0	0.11840	
1	20.57	17.77	132.90	1326.0	0.08474	
2	19.69	21.25	130.00	1203.0	0.10960	
3	11.42	20.38	77.58	386.1	0.14250	
4	20.29	14.34	135.10	1297.0	0.10030	

	mean compactness	mean concavity	mean concave points	mean symmetry	\
0	0.27760	0.3001	0.14710	0.2419	
1	0.07864	0.0869	0.07017	0.1812	
2	0.15990	0.1974	0.12790	0.2069	
3	0.28390	0.2414	0.10520	0.2597	
4	0.13280	0.1980	0.10430	0.1809	

	mean fractal dimension	...	worst texture	worst perimeter	worst area	\
0	0.07871	...	17.33	184.60	2019.0	
1	0.05667	...	23.41	158.80	1956.0	
2	0.05999	...	25.53	152.50	1709.0	
3	0.09744	...	26.50	98.87	567.7	
4	0.05883	...	16.67	152.20	1575.0	

	worst smoothness	worst compactness	worst concavity	worst concave points	\
0	0.1622	0.6656	0.7119	0.2654	
1	0.1238	0.1866	0.2416	0.1860	
2	0.1444	0.4245	0.4504	0.2430	
3	0.2098	0.8663	0.6869	0.2575	
4	0.1374	0.2050	0.4000	0.1625	

	worst symmetry	worst fractal dimension	target
0	0.4601	0.11890	0
1	0.2750	0.08902	0
2	0.3613	0.08758	0
3	0.6638	0.17300	0
4	0.2364	0.07678	0

[5 rows x 31 columns]

Training samples: 455

Testing samples: 114

===== MODEL EVALUATION =====
Accuracy: 0.9649122807017544

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.93	0.95	43
1	0.96	0.99	0.97	71
accuracy			0.96	114
macro avg	0.97	0.96	0.96	114
weighted avg	0.97	0.96	0.96	114

Confusion Matrix:

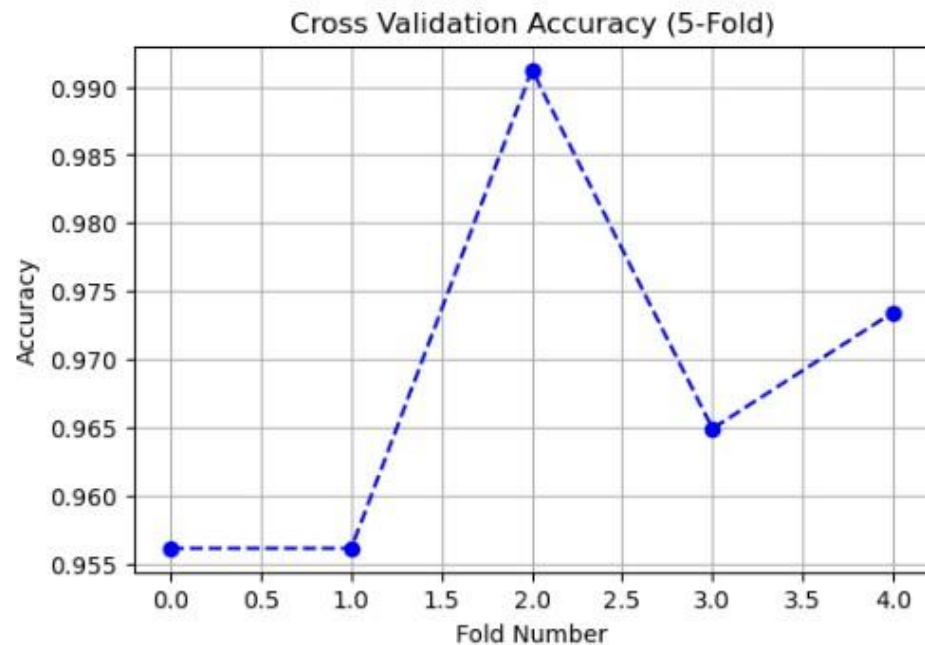
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[[40  3]
 [ 1 70]]
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===== CROSS VALIDATION =====

Fold Accuracies: [0.95614035 0.95614035 0.99122807 0.96491228 0.97345133]

Mean Accuracy: 0.9683744760130415

Standard Deviation: 0.013110267873227947



===== PARAMETER TUNING =====

n_estimators = 10, Mean Accuracy = 0.9456

n_estimators = 50, Mean Accuracy = 0.9684

n_estimators = 100, Mean Accuracy = 0.9772

n_estimators = 200, Mean Accuracy = 0.9772